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Section: Original Investigation

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"Reducing the Loss of Velocity and Power in Women Athletes via Rest Redistribution" by Merrigan JJ et al.
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Reducing the Loss of Velocity and Power in Women Athletes via Rest Redistribution

Original Investigation

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Abstract

Purpose: To examine rest-redistribution (RR) effects on back-squat kinetics and kinematics in resistance-trained women. **Methods**: Twelve women from strength and college sports (5.0 ± 2.2 y training history) participated in the randomized crossover-design study with 72 h between sessions (3 total). Participants completed 4 sets of 10 repetitions using traditional sets (TS, 120 s interset rest) and RR (30 s intraset rest in the middle of each set; 90 s interset rest) with 70% of their 1-repetition maximum. Kinetics and kinematics were sampled via force plate and 4 linear position transducers. The greatest value of repetitions 1–3 (peak repetition) was used to calculate percentage loss, [repetition 10 – peak repetition)/peak repetition) x 100], and maintenance, {100 – [(set mean – peak repetition)/peak repetition)] x 100}, of velocity and power for each set. Repeated-measures analysis of variance was used for analyses ($p < .05$). **Results**: Mean and peak force did not differ between conditions. A condition x repetition interaction existed for peak power ($p = .049$) but not peak velocity ($p = .110$). Peak power was greater in repetitions 7–9 ($p < .05$; $d = 1.12–1.27$) during RR. The percentage loss of velocity (CI95% –0.22, –7.22%; $p = .039$) and power (CI95% –1.53, –7.87%; $p = .008$) were reduced in RR. Mean velocity maintenance of sets 3 ($p = .036$, $d = 1.90$) and 4 ($p = .015$, $d = 2.30$) and mean power maintenance of set 4 ($p = .006$, $d = 2.65$) were greater in RR. **Conclusion**: By redistributing a portion of long interset rest into the middle of a set, velocity and power were better maintained. Therefore, redistributing rest may be beneficial for reducing fatigue in resistance-trained women.

Increasing power output is important when training athletes. Power is the product of force and velocity; therefore, changes in velocity are reciprocal for power.[1] During traditional resistance training, repetitions are performed continuously, without rest, until the set is complete. Within traditional sets, the velocity generally decreases,[2] and the magnitude of velocity loss is directly related to the number of sets and repetitions performed. For example, three sets of eight repetitions leads to a 32% reduction in velocity, whereas three sets of twelve repetitions leads to a 46% reduction in concentric velocity across sets.[3] Therefore, this style of acute training may not be ideal when aiming to maintain power, as has been demonstrated in the jump squat,[4,5] back squat,[1,6] and weightlifting exercises.[7,8] Of interest, is the manipulation of training variables, such as rest time, that may reduce the neuromuscular fatigue induced from high volume training protocols.

Anaerobic glycolysis is likely prevalent during high volume training (multiple high-repetition sets) and thus, fatigue may result from a reduction of immediate energy stores (adenosine triphosphate, ATP; phosphocreatine, PCr) and accumulation of lactate or ammonia.[9] Such accumulation may lead to early acidosis, indirectly interfere with cross-bridge formation, with the end result being reduced movement velocity.[3] Other findings refute acidosis as the grounds for fatigue.[10] Nevertheless, a rapid decrease in PCr stores results in transient increases of adenosine diphosphate and fatigue-induced decreases in muscle shortening velocities.[10] One strategy to combat such a series of events is to incorporate short intra-set rest periods,[11] which may permit energy systems to keep up with demand by dampening ATP and PCr depletion.[3,9] The use of intra-set rest will attenuate the reduction in velocity[7,8,12] and power output[1,8,9] that are generally noted in traditional sets. Further, a better understanding of the acute effects of intra-set rest may improve future program design.

In low-volume, power-based training, the addition of intra-set rest has allowed greater velocity, power, and force in the clean pull,[7] bench press,[13] and power clean.[8] Recently, similar results were reported in high-volume back squat training where intra-set rest, via rest redistribution (intra-set rest from part of total inter-set rest) or cluster sets (intra-set rest from additional rest time), reduced the loss of velocity and power.[1,12] However, the majority of the intra-set rest studies involved men, bringing into question whether or not women would respond in a similar fashion. Men consistently have more muscle mass than women, which can result in increased intramuscular pressure, impaired oxygen delivery, and increased metabolite accumulation.[14,15] Men also have greater type II muscle fiber area which may allow for faster contraction speeds and thus, faster rates of energy utilization and slower recovery between movement bouts.[15–17] Women may be more resistant to fatigue during training,[18] yet little is known about the impact of intra-set rest during resistance training in women. Recent evidence, has suggested inter-repetition rest to result in higher training velocities[19] or volumes[20] when compared to traditional sets in the resistance-trained. However, the aforementioned studies utilized either heavy loads or fatiguing protocols to failure and did not address high-volume training without failure. Therefore, the purpose was to examine the acute effects of rest redistribution and traditional sets on back squat kinetics and kinematics in resistance-trained women.

METHODS

Subjects

Twelve resistance-trained, competitive athletes from strength and collegiate sports (lacrosse, n=2; powerlifting, n=4; Olympic lifting, n=4; Crossfit, n=2), participated (Table 1). Subjects had >2 years of experience with the back squat and were currently participating in a structured resistance training program that included the back squat. Inclusion/exclusion criteria included: (1) women between 18 and 35 years of age; (2) non-pregnant; (3) not taking androgenic medications; (4) no lower body musculoskeletal injury in the past 6 months. Subjects were not participating in a weight class event or physique competition and were not in an intended caloric restriction. This study was conducted according to the Declaration of Helsinki guidelines and all procedures were approved by the University's Institutional Review Board for use of human subjects in research. Written consent was obtained from subjects prior to data collection.

Design

To evaluate the kinetic and kinematic differences between rest redistribution (RR) and traditional sets (TS), a repeated-measures, counterbalanced design was employed. Each subject served as her own control by completing both experimental conditions (RR, TS). Sessions took place during the follicular phase of each subject's menstrual cycle. After measuring height, weight, and body composition, subjects completed a 10-minute dynamic warm-up and demonstrated proficiency in the back squat with a barbell. Subjects were asked to pause at the bottom of a squat to ensure they could reach the desired depth (top of the thigh parallel to the floor). Once proficiency was confirmed by the same certified strength and conditioning specialist, subjects continued an incremental squat protocol to determine their one-repetition maximum (1-RM). At the end of the final rep, foot placement and safety bar height were measured and recorded for use in subsequent trials. Subjects were asked to wear the same footwear to all sessions. Seventy-two hours later, subjects completed the two experimental sessions in a counterbalanced order, separated by 72 hours.

Body Composition

Upon arrival to the laboratory, height and body mass were recorded to the nearest 0.01 cm and 0.02 kg, respectively using a stadiometer (Detecto, Webb City, MO) and digital scale (BOD

POD; Cosmed USA, Concord, CA) with subjects bare foot. Body composition was assessed using air displacement plethysmography (BOD POD model 2000A; BOD POD; Cosmed USA, Concord, CA). Prior to each testing session, calibration procedures were completed. Subjects wore a formfitting sports bra, spandex shorts, swim cap, and removed all jewelry, in accordance with standard operating procedures. Subjects entered the BOD POD and sat still in an erect position for the duration of the measurement. Intraclass correlation coefficients were; body mass, r=1.000, body fat percentage, r=0.997; and fat free mass r=1.000.

One‑repetition Maximum Testing and Familiarization

Following body composition testing, subjects began 1-RM testing in the back squat. To ensure safety and decrease injury risk, each subject underwent a supervised and standardized whole body warm up before back squat testing, which consisted of a 5-minute treadmill walk followed by 10 minutes of dynamic stretching. After the warm up, and under researcher supervision, subjects performed two sets of five repetitions at 40–60 % of their self-estimated 1-RM with 2 minutes rest between sets. Following a 3-minute rest, subjects completed 1 set of 2-3 repetitions at 80% self-estimated 1-RM. Next, single repetitions were performed with 3-5 minutes of rest between attempts, and the load was progressively increased until the subject reached her true 1-RM, failed two consecutive attempts, or could not maintain the proper form or depth, thereby concluding the 1-RM session. A valid attempt consisted of the top of the thigh parallel to the floor, which was visually assessed by the same trained researcher and supplemented with auditory signaling from a squat beeper (Bigger, Faster, Stronger; Safety Squat). All 1-RM determinations were made within 3–5 attempts with safety bars in place. This procedure for establishing the 1-RM back squat has been shown to have an intraclass correlation coefficient of 0.99.[21]

Experimental Testing

On the day of experimental testing, subjects arrived having refrained from lower body training for at least 72 hours and any activities outside of daily living for the previous 48 hours. After the supervised 15-minute warm-up, identical to that performed for 1-RM determination, subjects performed two sets of five repetitions of the parallel back squat exercise with a load equivalent to 40 and 60 % of 1-RM. Following two minutes of rest, subjects performed the back squat using 70% 1-RM for their respective protocol (TS, RR) for that day. For the TS protocol, subjects completed 4 sets of 10 repetitions with 120 seconds of inter-set rest, and for the RR protocol, subjects performed 4 sets of 10 repetitions with 90 seconds on inter-set rest and 30 seconds of intra-set rest after the 5th repetition of each set (Figure 1). This procedure was fatiguing, but did not lead to failure. All subjects completed a total of 40 back squat repetitions in both conditions. Subjects remained seated during the inter-set rest and standing during intra-set rest. Subjects were instructed to perform each repetition "as explosively as possible" with verbal encouragement throughout all experimental testing to help ensure maximal effort on each repetition.

Data Acquisition and Analysis

Subjects performed experimental testing on a force platform (AccuPower, Advanced Mechanical Technology Inc., AMTI; Watertown, MA, USA) with four linear position transducers (LPT) (PT5A-150 Celesco, Measurement Specialties, Chatsworth, CA, USA) attached to the bar on the inside of both collars. The cables were mounted above and anterior and above and posterior, to the subject, to form two triangles when attached to the barbell, which allowed for the measurement of horizontal and vertical bar displacement [22]. The LPTs produced a voltage signal that quantified the displacement and time of LPT movement. The displacement-time data were single differentiated to provide instantaneous velocity measures throughout the movement. Ground reaction force, calculated via force plate, and displacement data were sampled at a frequency of 1,600 Hz using an interface box with an analog-to-digital card. Signals were filtered using a second-order Butterworth low-pass filter with a cutoff frequency of 20 Hz (NI cDAQ-9174; National Instruments, Austin, TX, USA). The movement was initiated when a velocity or displacement threshold was > 0.01 $m \cdot s^{-1}$ or 0.01 m, respectively, and terminated when velocity decreased to < 0.01 $m \cdot s^{-1}$. The force plate was calibrated prior to subjects stepping onto it. The force calculation included the mass of the subject and bar (weight of total load + total load multiplied by the acceleration of the system's center of mass). Power was calculated as the product of net force and velocity for each repetition over all sets. LabView full development system (Version 8.2 National Instruments, Austin, TX, USA) was used to collect and calculate data from the devices to output kinetics and kinematics. Peak and average values during the concentric phase were then obtained for comparison. The greatest value of repetitions 1-3 (peak repetition) was used to calculate percent loss [rep10 – peak repetition) / peak repetition) x 100] and maintenance [100 – ((set mean – peak repetition)/peak repetition)) x 100] of velocity and power for each set. Further explanation of these calculations is presented in previous literature.[3,23] Intraclass correlation coefficients for concentric velocity, power, and force were r=0.93, r=0.98, and r=0.97, respectively.

Statistical Analysis

Data were normally distributed as determined by the Kolmogorov–Smirnov test. Means and standard deviations were used to represent centrality and spread of data, unless otherwise noted. A priori power analysis (G*Power version 3.1.9) determined the minimum sample size required to find significance was equal to 8 subjects. The analysis was run with a desired level of power equal to 0.80, an α-level set at 0.05, and a standardized effect size (d=0.4) calculated from previous studies.[1,12,23] Multifactorial analysis of variance with repeated measures was used to analyze data with the following factors: condition (2 levels), set (4 levels), and repetition (10 levels). Bonferroni post hoc analysis was performed when a significant finding ($p \leq 0.05$) was identified. All analyses were performed using SPSS V.24 (IBM Corporation; Armonk, NY). Effect sizes were calculated using Cohen's d and are defined as small, d = 0.2; medium, d = 0.5; and large, d = 0.8.[24]

RESULTS

There were no main condition effects nor interactions for mean (p=0.122, η_p^2=0.128, Table 2) and peak force (p=0.301, η_p^2=0.099). Significant main effects and interactions are displayed in the text, whereas specific p-values and effect sizes are presented in each respective figure and table. Mean and peak velocity and power of repetitions, collapsed across sets, are presented in Figure 2. Mean percent loss and maintenance of velocity and power across sets are presented in Figure 3. Concentric velocity and power of each repetition across sets of TS and RR conditions are displayed in Figure 4.

There was no main condition effect for velocity (p=0.930, η_p^2=0.001). Condition*repetition interactions existed for mean velocity (p=0.002, η_p^2=0.896), but not peak velocity (p=0.110, η_p^2=0.936). There was a condition (p=0.039, η_p^2=0.332) and set (p=0.017, η_p^2=0.264) effect for mean velocity loss across sets, but no interaction existed (p=0.756, η_p^2=0.035). Maintenance of mean and peak velocity differed between conditions (p=0.004, η_p^2=0.542). No condition by set interactions existed for maintenance variables.

For power, no main condition effect existed (p=0.327, η_p^2=0.087). Despite large effects, there were no condition*repetition interactions for mean power (p=0.161, η_p^2=0.915) and peak power (p=0.281, η_p^2=0.933). Subjects demonstrated less power in later repetitions during TS compared with RR (Figure 2). Main condition (p=0.008, η_p^2=0.492) and set (p=0.012, η_p^2=0.278) effects existed for mean power loss. Maintenance of mean (p=0.002, η_p^2=0.595) and peak power (p=0.006, η_p^2=0.506) was different between conditions. No condition by set interactions existed for percent loss or maintenance for mean or peak power (p>0.05).

DISCUSSION

Since women may be more resistant to fatigue during training,[18] the current study was conducted with similar RR methodology as previously researched in men.[1] The hypothesis was that velocity and power would be attenuated through the use of RR in high-repetition back squat sets. Although, there were no main condition effects for velocity and power, select repetitions late in sets of RR were higher than TS. Further, the main finding was that the utilization of RR reduced velocity and power loss within sets compared to TS.

In some instances, the objective of resistance training may include simultaneously focusing upon strength, hypertrophy, and power development. Further, the combination of strength and power training seems to enhance performances in strength and power tasks for women.[25] However, since muscle size is a main contributor to strength, a multifaceted approach to resistance training is often warranted. Although high volumes of resistance training are suitable for muscle hypertrophy, they may elicit fatigue that subsequently hinders mechanical power output and movement velocity. Yet, with proper distribution of rest, it is possible to have similar workloads in training without sacrificing velocity and power,[1,12,23] leading to similar hypertrophic changes with greater increases in strength and power.[21] Therefore, it is important to understand if the use of intra-set rest has similar benefits for kinetics and kinematics in women as men.

In the current study, mean and peak forces were no different between RR and TS, which is in agreement with previous findings in men.[1,12,26] When increasing total rest rather than simply redistributing rest the outcome may differ. For example, inclusion of an additional 40 seconds of inter-repetition rest, and thus greater total rest, resulted in improved maintenance of force and velocity during the power clean exercise.[8] In addition, exercises that have inconsistent bar displacement may require varying forces to complete the movement. During the back squat, all repetitions have nearly identical bar displacement and should yield consistent force outcomes between conditions, as long as the external load remains constant during each protocol.[12,26,27] Oliver et al. reported lower force during traditional sets in the 1st repetition and later repetitions (6-10) when collapsed across sets, which was the result of the male participants’ inability to complete all repetitions.[1] In the current study, under the same protocols, all women were able to complete all repetitions for each set; therefore, the load was the same in each condition resulting in similar forces.

When force is similar between conditions, velocity is the main contributor to variations in a movement’s power.[1,12] However, in the current study a condition effect existed solely for mean velocity, which may be a result of lower between subject variability compared to that of mean power. In a prior study with men, the use of similar intra-set rest allowed subjects to begin later sets with higher velocities compared to traditional sets, which was likely due to the lighter loads utilized in later sets.[1] The velocity of initial repetitions, in the current study, was lower in RR than TS (Figures 2 and 4). The lower velocity of initial repetitions when returning from intra-set rest has been previously noted[27] and may explain why the acute kinetic and kinematic responses were similar overall, regardless of inclusion of intra-set rest. When returning from a designated rest (52.5 or 120 sec), the initial repetition produced the lowest peak force, compared to the 3 subsequent repetitions.[27] In the RR condition of the current study, 30 seconds of intra-set rest was redistributed from inter-set rest, therefore, the longer inter-set rest during TS enhanced the dissipation of fatigue.

When total rest is identical between conditions, and TS protocols are not fatiguing, velocity and power do not decrease from set to set in RR and TS conditions.[27,28] In previous studies with men, fatiguing traditional sets led to decreased velocity and power in later sets.[1,3,12] Further, when inter-set rest remained constant, the longest intra-set rest provided the greatest velocity and power output.[23] Similar findings were noted with inter-repetition rest, as the addition of either 15 or 30 seconds of inter-repetition rest resulted in higher maintenance of velocity within sets of 10 repetitions.[29] With increased total rest time the phosphogen and glycolytic energy pathways are better preserved and result in greater power performances.[30,31] Therefore, it is possible that the inclusion of intra-set rest may not result in higher absolute values of velocity and power in later repetitions, compared to traditional sets, unless the rest is added instead of redistributed.

In the current study, velocity and power losses were lower in RR providing ~20% loss during the first 3 sets and 24% during set 4. In the TS condition, women subjects experienced a 22% loss in mean velocity and power, which increased proportionately to about 32% from set 1 to set 4. Prior literature reported similar losses of 20-23% in the back squat,[1,12] 37% in the leg press,[9] 18.5% in the power clean.[8] In the same studies velocity and power losses were between 1-5% during back squat cluster sets (3 sets of 12 repetitions at 60% 1-RM) with 30 seconds of intra-set rest[12] and 5-8.5% when 20-40 seconds of inter-repetition rest was used during 80% 1-RM power cleans.[8] A possible explanation for the differences is that the current study maintained total rest time of 120 seconds, with 30 seconds redistributed to intra-set rest, while others kept inter-set rest constant at 120 seconds and included additional rest within each set.[12] Previous literature, using equated total rest during 4 sets of 6 repetitions of 40 kg jump squats, reported approximately 12% power loss and 8% velocity loss from repetition 1 to 6 in traditional and 2-6% power loss and less than 3% velocity loss when using some type of cluster set.[4] Although the velocity loss in the current study was not as low, the relative difference in losses between RR and TS was similar, despite different exercises. Therefore, additional rest within a set may have greater effects when total rest time is increased as opposed to redistributing pre-existing inter-set rest. Nonetheless, the velocity loss within sets may still be attenuated from redistributing rest.

The decreased rest time between sets during the RR condition may not have promoted adequate recovery of the ATP and PCR system, resulting in lower velocity and power of initial repetitions. Compared to men, women have fewer type II fibers in the vastus lateralis[32], which may have attenuated the extent of fatigue accumulated during traditional sets. Women also may have slower declines in force generation, longer time to exhaustion, and less impairment to neuromuscular activation following fatiguing exercise,[33,34] which may explain some of the divergent findings between the women of the present study and men in previously published research utilizing similar protocols. However, when training with heavy loads in 6 sets of 4 repetitions with the bench press exercise, men and women demonstrated similar velocity losses in traditional sets and rest redistribution.[19] In the same study, the use of inter-repetition rest resulted in higher velocities during the last repetition of each set compared to sets using traditional styles and rest redistribution. Therefore, potential differences between men and women in their response to rest redistribution may be dependent upon the exercise and the set and rep scheme.

Practical Applications

Redistributing part of an inter-set rest interval into the middle of a 10-repetition set may improve maintenance of velocity and power compared to traditional sets of high-volume back squats in resistance-trained women. Since no absolute differences in velocity and power were observed, the redistribution of rest may not be beneficial in producing higher absolute power output, but rather serve to reduce fatigue. Had there been additional intra-set rest (as used in cluster set protocols), rather than redistributing the inter-set rest, the patterns noted in absolute velocity and power may have increased during RR. Even so, these data support the use of rest redistribution in resistance training when reductions in fatigue are desired.

Conclusions

Overall, redistributing 30 seconds of inter-set rest to the middle of a set resulted in better maintenance of velocity and power in resistance-trained women. Therefore, redistributing rest may attenuate fatigue-induced decreases in velocity, as demonstrated by the greater maintenance of velocity within certain sets.

Acknowledgements

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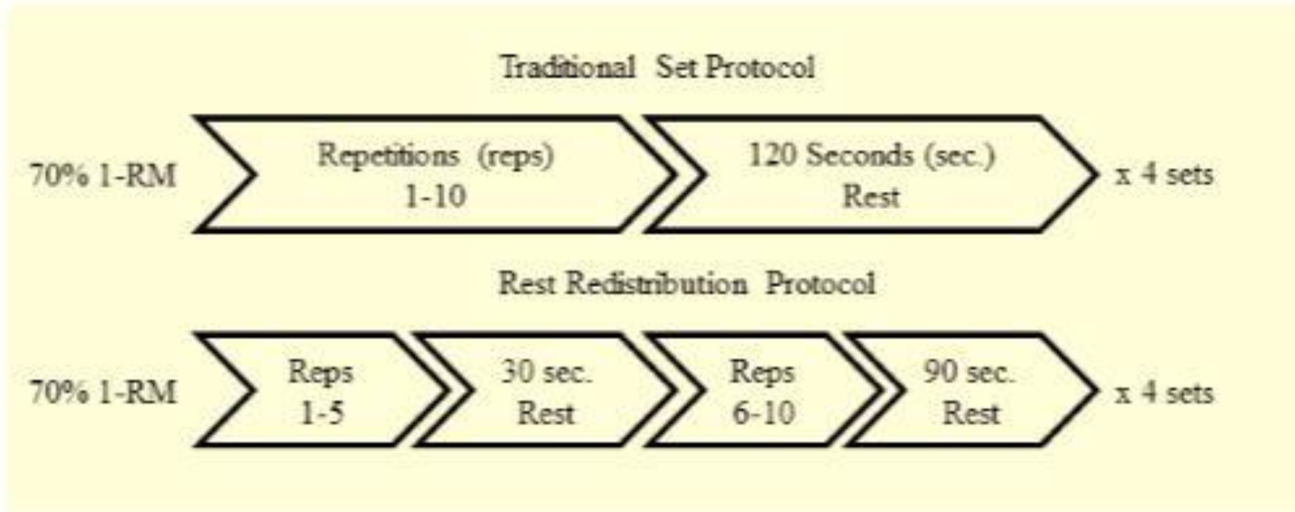

Figure 1. Configuration of each condition protocol.

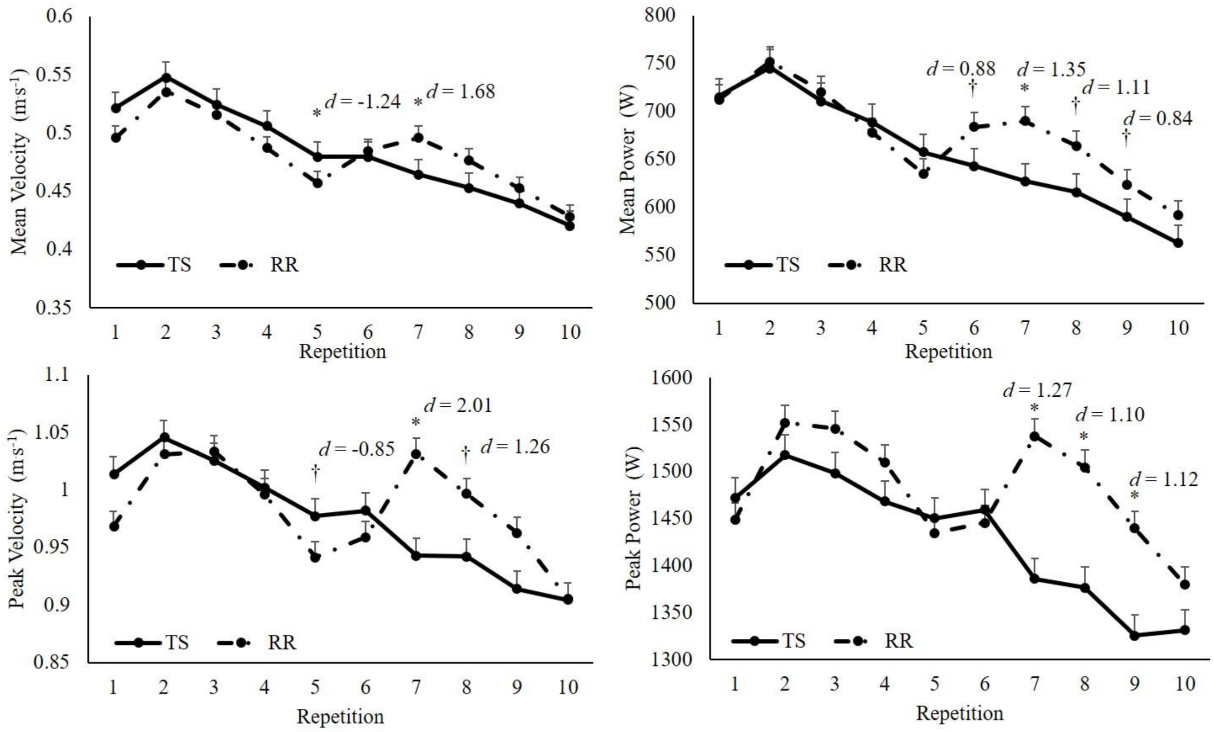

Figure 2. Mean and peak velocity and power of each repetition collapsed across sets of back squats. *, Significant difference between rest redistribution (RR) and traditional sets (TS) ($p < 0.05$); † denotes a trend towards significance ($p < 0.10$). The effect size is listed for each statistically significant comparison and trend.

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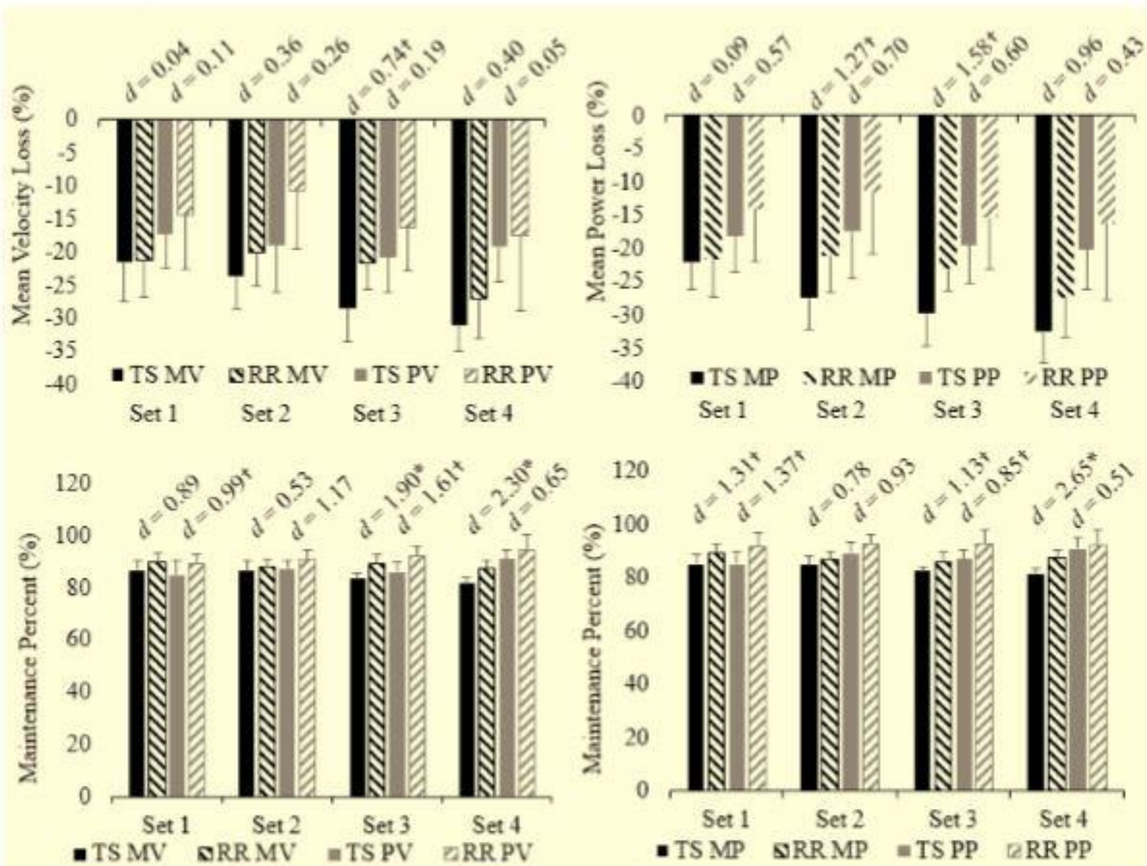

Figure 3. Percent loss of mean and peak velocity (MV, PV) and power (MP, PP), as well as maintenance of each variable, across repetitions throughout each set of back squats. *, Significant difference between rest redistribution (RR) and traditional sets (TS) ($p < 0.05$); † denotes a trend towards significance ($p < 0.10$). The effect size is listed for each comparison.

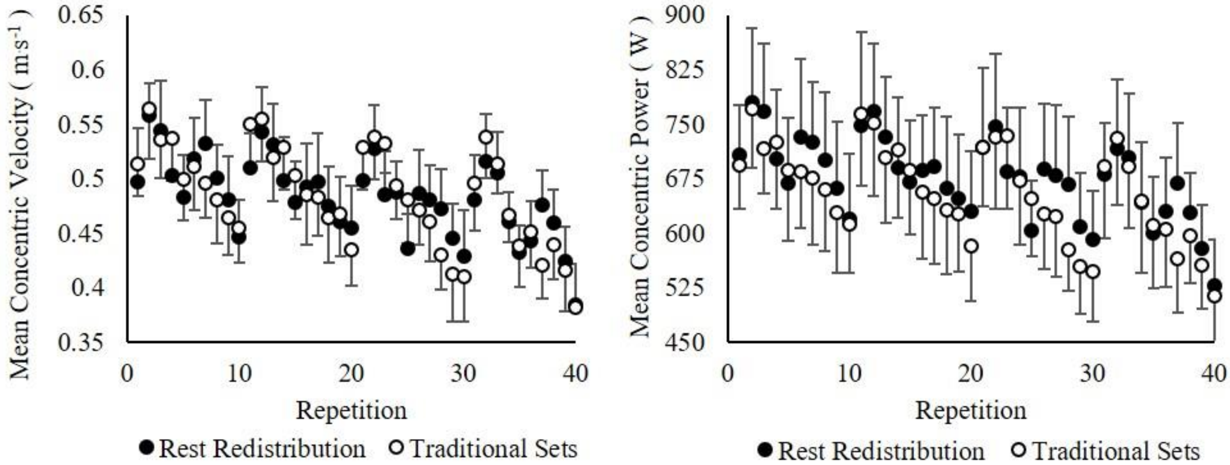

Figure 4. Layout of mean concentric velocity and power for each repetition performed in rest redistribution (solid circles) and traditional sets (non-filled circles).

"Reducing the Loss of Velocity and Power in Women Athletes via Rest Redistribution" by Merrigan JJ et al.
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Table 1: Descriptive data of resistance-trained women.

Variable	Mean ± SD
Age (years)	23.7 ± 4.1
Height (cm)	159.8 ± 5.6
Body Mass (kg)	64.1 ± 10.8
Body Fat (%)	24.2 ± 4.6
Training Age (years)	5.0 ± 2.2
One Repetition Maximum (1-RM, kg)	101.7 ± 20.2
1-RM Back Squat: Body Mass	1.6 ± 0.2